

# Math 571 - Homework 3

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Problems numbered [B:2.1:3] are from [Bergman's exercise notes](#). Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (\*) to a problem, we have a slight variant of the problem.

**Notation:**  $\{p_i\} = \{p_i \mid i = 1, 2, \dots\} \subseteq X$  is a set, which may or may not be infinite.  $(p_i) = (p_i \mid i = 1, 2, \dots) = (p_1, p_2, \dots) \in X^{\mathbb{N}}$  is an infinite sequence from  $X$ . We might start counting from 0 instead of 1 on occasion, and if we need to start from another point, we make this explicit, using notation like  $\{p_i\}_{i=k}^{\infty}$  and  $(p_i)_{i=k}^{\infty}$ . It is important to have these two notations, for example, if  $p_i = (-1)^i$ , then  $\{p_i\} = \{-1, 1\}$  while  $(p_i) = (-1, 1, -1, 1, \dots)$  and if  $q_i = (-1)^{i+1}$ , then  $\{q_i\} = \{p_i\} = \{-1, 1\}$  while  $(q_i) = (1, -1, 1, -1, \dots) = -(p_i)$ .

**Problem 3.1** (B:3.1:1 - A convergent sequence together with the point it approaches form a compact set.). Let  $X$  be a metric space.

- (a) Show that if  $(p_i)$  is a sequence from  $X$  with  $p = \lim_{i \rightarrow \infty} p_i = p$ , then  $\{p\} \cup \{p_i\}$  is compact.
- (b) Deduce that for all  $S \subseteq X$ ,  $S$  is closed iff for all compact  $K \subseteq X$ ,  $S \cap K$  is closed.

**Problem 3.2** (B:3.2.1 - Converse to Theorem 3.6 (a).). Let  $X$  be a metric space. Show that if every sequence  $(x_i)$  in  $X$  has a convergent subsequence, then  $X$  is compact.

**Hint:** You can use R:2.26 which was homework problem 2.7.

**Note:** The statement: Every sequence has a convergent subsequence is called *sequential compactness*. So this problem has you show that if  $X$  is metric, then:  $X$  is sequentially compact iff  $X$  is compact.

**Problem 3.3** (R:3.20). Let  $X$  be a metric space and  $(p_i)$  be a Cauchy sequence with a convergent subsequence. Show that  $(p_i)$  converges.

**Problem 3.4** (R:3.22 - Baire Category Theorem). Suppose  $X$  is a complete metric space and  $\{G_n\}$  is a set of dense open subsets. Show that  $\bigcap G_n$  is dense.

**Problem 3.5** (Application of Baire Category).

- (a) Show that  $\mathbb{Q}$  is a countable union of closed sets.
- (b) Show that  $\mathbb{Q}$  is not a countable intersection of open sets.

**Problem 3.6** (R:3:16(18)\*). Fix  $\alpha > 1$  and  $x_1 > \sqrt{\alpha}$ , define

$$x_{n+1} = \frac{1}{2} \left( x_n + \frac{\alpha}{x_n} \right)$$

- a) Prove that  $(x_n)$  decreases monotonically and  $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$ .
- b) Let  $\varepsilon_n = x_n - \sqrt{\alpha}$  be the error at the  $n^{\text{th}}$  term, show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$$

Setting  $\beta = 2\sqrt{\alpha}$ , gives

$$\varepsilon_{n+1} < \beta \left( \frac{\varepsilon_1}{\beta} \right)^{2^n}$$

- c) Choose a number  $\alpha > 3$  find a bound for how many terms are need to compute  $\sqrt{\alpha}$  correct to 20 decimal places where  $x_1$  is chosen minimally so that  $x_1^2 > \alpha$ . Prove your answer and do the computation. You might use [Python](#) or MATLAB.
- d) Replace the recursion above by

$$x_{n+1} = \frac{p-1}{p} x_n + \frac{\alpha}{p} x_n^{-p+1}$$

Discuss the behavior of  $(x_n)$  under suitable conditions. Don't bother trying to compute a recursive expression for  $\varepsilon_n$  in this case, but do prove your claims.

**Problem 3.7** (B:3.7:1 - No boundary between convergent and divergent series.). The statement Rudin makes on p.63 that there is no “boundary” between convergent and divergent series can be made precise as follows: Given sequences  $(a_n)$ ,  $(b_n)$  of positive real numbers, each having limit 0, let us say that “ $(a_n)$  decays more rapidly than  $(b_n)$ ”, equivalently that “ $(b_n)$  decays more slowly than  $(a_n)$ ” if  $\lim_{n \rightarrow \infty} a_n/b_n = 0$ , equivalently,  $\lim_{n \rightarrow \infty} b_n/a_n = +\infty$ . We note that if a sequence of positive terms has a convergent sum, so does every sequence of positive terms that decays more rapidly; and if a sequence of positive terms has a divergent sum, then so does every sequence of positive terms that decays more slowly.

Let us call  $(a_n)$  a “boundary sequence” if every sequence that decays more rapidly than  $(a_n)$  converges and every sequence that decays more slowly than  $(a_n)$  diverges. Rudin's **3:R11** and **3:R12** show respectively that given a divergent series  $\sum a_n$ , one can find a sequence  $(b_n)$  which decays more rapidly than  $(a_n)$  but such that  $\sum b_n$  still diverges, and that given a convergent series  $\sum a_n$ , one can find a sequence  $(b_n)$  which decays more slowly than  $(a_n)$  but such that  $\sum b_n$  still converges.

- (a) Deduce from the above facts that no series  $\sum a_n$ , whether divergent or convergent, can satisfy the definition given above for a “boundary sequence” (thus justifying Rudin's claim). However, the constructions of **3:R11** and **3:R12** are unnecessarily complicated. Here is a simpler pair of constructions:
- (b) Suppose  $\sum a_n$  is a divergent series of positive terms with  $\lim_{n \rightarrow \infty} a_n = 0$ . Show that there is an increasing sequence  $(A_n)$  of positive real numbers with  $\lim_{n \rightarrow \infty} A_n = +\infty$  such that  $A_{n+1} - A_n = a_n$  for all  $n$ . Now let  $B_n = A_n^{1/2}$ , and define  $b_n = B_{n+1} - B_n$ . Show that  $(b_n)$

is also a sequence of positive terms that approaches 0, and that it decays more rapidly than  $(a_n)$ , but that  $(B_n)$  still approaches  $+\infty$ , hence that  $\sum b_n$  is still divergent. (Hint: Here and in part (b), the relation  $B_n = A_n^{1/2}$  is more easily used in the form  $A_n = B_n^2$ ).

- (c) Suppose  $\sum a_n$  is a convergent series of positive terms. Show that there is a decreasing sequence  $(A_n)$  of positive real numbers with  $\lim_{n \rightarrow \infty} A_n = 0$  such that  $A_n - A_{n+1} = a_n$  for all  $n$ . Again let  $B_n = A_n^{1/2}$  and now let  $b_n = B_n - B_{n+1}$ . Show that  $(B_n)$  also approaches 0, hence that  $\sum b_n$  converges, but that  $(b_n)$  decays less rapidly than  $(a_n)$ .

**Problem 3.8** (R:3.9). Find the radius of convergence and **region interval of convergence** of the following power series. If you take  $z \in \mathbb{R}$ , then the region of convergence is an interval, but if  $z \in \mathbb{C}$ , then this region is the open disk  $N_R(0)$  together with part of its boundary. The complex region might be difficult to calculate, but give it a try for the challenge. In all but one, the real interval is simple to calculate.

- a)  $\sum_n n^3 z^n$
- b)  $\sum_n \frac{2^n}{n!} z^n$
- c)  $\sum_n \frac{2^n}{n^2} z^n$
- d)  $\sum_n \frac{n^3}{3^n} z^n$
- e)  $\sum_n \frac{n^n}{n!} z^n$

**Problem 3.9** (B:3.13:1 - Radius of Convergence of sum and product series.). Let  $\sum_n a_n$  and  $\sum_n b_n$  be two series and let  $\sum_n a_n + \sum_n b_n = \sum_n s_n$  and  $(\sum_n a_n)(\sum_n b_n) = \sum_n p_n$ . Recall that this means  $s_n = a_n + b_n$  while  $s_n = \sum_{i+j=n} a_i b_j = \sum_{i=1}^n a_i b_{n-i}$ . Let  $R_a$ ,  $R_b$ ,  $R_s$ , and  $R_p$  be the radius of convergence of  $\sum_n a_n$ ,  $\sum_n b_n$ ,  $\sum_n s_n$ , and  $\sum_n p_n$  respectively.

- (a) Show that  $\min\{R_a, R_b\} \leq \min\{R_s, R_p\}$ .
- (b) Deduce that if  $R_a < R_b$ , then  $R_s = R_a$ . What might happen if  $R_a = R_b$ ?
- (c) Deduce that if  $R_p < R_a$ , then  $R_b \leq R_p$ .

**Problem 3.10** (B:3.14:2 - Which series have convergent rearrangements?). Find a simple and easy-to-check criterion for a series  $\sum a_n$  to have a rearrangement  $\sum a_{\sigma(n)}$  that converges, for some  $\sigma$  a permutation of  $\mathbb{Z}^+$ .